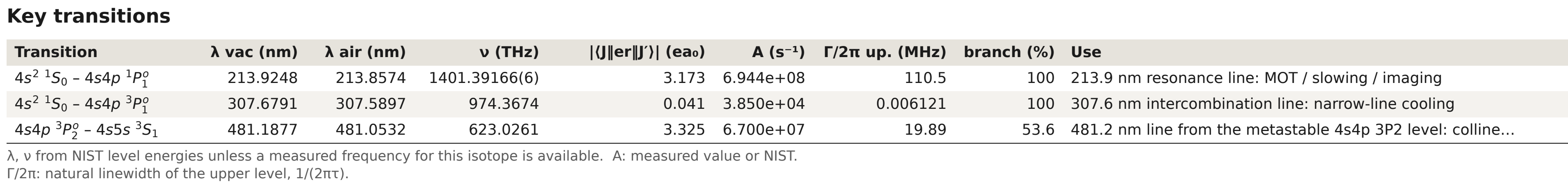


Zn (Zn I). Strongest classified transitions below 2 μm (NIST): 47 levels, 70 lines.

Zn⁺ ionisation limit 75769.31 cm⁻¹ = 9.39420 eV (131.980 nm)



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

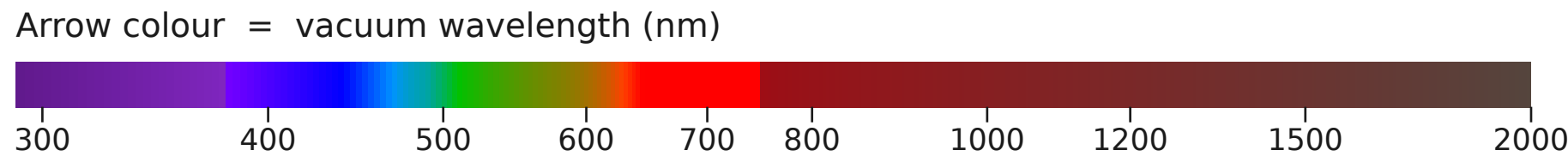
$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Arrow label: vacuum wavelength λ reduced dipole matrix element $| \langle l \| e r \| l' \rangle |$ in each

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



⁶⁷Zn hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
4s4p $3P_0^o$	608.922(3)	-18.995(10)	F=3/2: -2144.52 F=5/2: -593.73 F=7/2: +1517.56
4s4p $3P_2^o$	531.987(5)	35.806(5)	F=1/2: -3698.84 F=3/2: -2916.98 F=5/2: -1604.91 F=7/2: +250.78 F=9/2: +2668.89
4s4p $1P_1^o$	20.0(20)	10.5(5)	F=3/2: -63.00 F=5/2: -28.00 F=7/2: +12.50
4s5s $3S_1$	1266.5(18)	-	F=3/2: -4432.75 F=5/2: -1266.50 F=7/2: +3166.25

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	213.925	66-64	525(3)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	213.925	67-64	835(4)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	213.925	68-64	1039.81(7)
$4s^2\ ^1S_0 - 4s4p\ ^1P_1^o$	213.925	70-64	1495(4)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	307.679	66-64	689.411(4)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	307.679	67-64	1085.933(7)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	307.679	68-64	1362.390(3)
$4s^2\ ^1S_0 - 4s4p\ ^3P_1^o$	307.679	70-64	1958.358(7)
$4s4p\ ^3P_2^o - 4s5s\ ^3S_1$	481.188	64-68	-141.2(12)
$4s4p\ ^3P_2^o - 4s5s\ ^3S_1$	481.188	66-68	-63.6(15)
$4s4p\ ^3P_2^o - 4s5s\ ^3S_1$	481.188	67-68	-41.4(2.1)
$4s4p\ ^3P_2^o - 4s5s\ ^3S_1$	481.188	70-68	69.5(9)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$